Uniqueness of the Classical Langton Ant Highway Under Its Standard Wake

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Abstract

The permanent wake of a periodic Langton ant can determine its motion uniquely. We prove that with drift $(2, -2)$ and the twelve complete odd translation classes of the classical highway, every finite-support periodic realization is the standard 104-step highway, up to phase and pose. The period is determined by the wake rather than supplied as a hypothesis. The proof combines a finite graph of necessary short-factor constraints with an exact ordered-event representation. An integer potential on the first graph bounds every candidate period by 168; exhaustive chronological matching on the second leaves one normalized word. Consecutive-visit ranks and separation of fixed trajectory chains reduce the verified enumeration from 40,227,976 to 8,580,751 prefixes. More generally, we retain wake-dependent period bounds for every non-axis drift, giving a terminating inverse-synthesis algorithm and an obstruction to wake-preserving detours. All finite certificates can be regenerated, and the accepted word is checked on an explicit finite seed for its entire future.

Keywords. Langton's ant; periodic highway; finite support; symbolic dynamics; computer-assisted proof; inverse synthesis.

1 Introduction

A periodic Langton ant leaves a geometric record of its motion: the permanent black strands in its wake. The classical highway repeats after 104 updates, translating by $(2, -2)$ in a suitable orientation. Its wake records which translation classes have been visited an odd number of times, while the motion also specifies the order of those visits. This paper proves that the complete standard wake determines that order uniquely among finite-support periodic realizations with the same drift.

The inverse problem starts with the wake and seeks the periodic motion that produces it. Its first difficulty is that the period is unknown. Its second is chronological: each visit flips a cell, so visits to the same physical cell must alternate right and left turns. Counts of visits give useful necessary conditions, but the order links different phases and different periods. Recovering the highway requires both a period budget and exact control of these links.

Main result. For the twelve standard classes O_{std} specified in Section 2, every heading-restoring periodic realization from a finite black set, with drift $(2, -2)$ and wake O_{std}, is the standard 104-step highway. A periodic suffix satisfying these hypotheses therefore has the standard future trace, with

any stationary unread cells retained. The theorem classifies motion under its complete wake; its hypotheses are stated directly in terms of the drift and translation classes.

Two complementary reductions make the classification finite. Short factors that contradict the literal colour-update rule define a finite context graph. Every candidate trajectory projects to a closed walk in this graph. An integer potential bounds its period by 168. We then enumerate ordered visits within the complete translation classes, retaining longitudinal residues and the chronological order across periods. This second representation is exact, and its completed enumeration leaves a single word. Consecutive-rank and chain-separation lemmas strengthen the search while preserving its exact coverage.

The uniqueness theorem completes the standard-wake problem posed in the first version of this preprint, *Reading the Wake* [5]. The general period bounds and inverse-synthesis theorem from that version remain in Appendix A. For every prescribed finite wake, non-axis drift and initial heading, a computable cutoff makes the inverse problem decidable and constructs a finite seed whenever one exists. The present theorem carries that programme through for the standard wake.

Related work. Gajardo studies Langton's ant through symbolic dynamics and forbidden factors [1]. Jillhewar gives an exact realization criterion for a supplied periodic word, a constructive finite seed, and structural restrictions on finite-support highways [2]. His width-four analysis also gives a special period–drift bound [3]. We recall the realization criterion with its proof, then develop the integer upper certificate and complete chronological classification for O_{std}. The general wake budgets use the integer clock from the companion escape paper [4]. These dependencies separate the supplied-word test, the quantitative period bounds, and the standard-wake uniqueness proved here.

Section 3 establishes the normalization and realization criterion. Sections 4–6 give the upper bound, exact event representation, and completed classification. Section 7 describes the independent all-future checks. The appendices retain the general synthesis theory, the explicit witness, and the reconstruction specifications.

2 Model, normalization, and the standard wake

A complete state is $X = (B, p, h)$, where $B \subset \mathbb{Z}^2$ is finite, $p \in \mathbb{Z}^2$, and $h \in \{\mathrm{N}, \mathrm{E}, \mathrm{S}, \mathrm{W}\}$. All cells outside B are white. With headings encoded by $0, 1, 2, 3$ in that order, the update F is

$$h' = h + 1 - 2\mathbf{1}_B(p) \pmod 4, \qquad B' = B \triangle \{p\}, \qquad p' = p + e_{h'}.$$

Time i denotes the state before update i. Write R for reading white and turning right, and L for reading black and turning left. Every assertion below uses this convention.

A word $w = w_0 \cdots w_{P-1}$ gives a prescribed walk, with pre-update positions q_i and headings h_i. A *periodic realization* is a complete finite state whose entire future turn stream is w^ω. We require $h_P = h_0$ and nonzero drift $d = q_P - q_0$. The full black field includes the growing wake and any stationary unread cells.

Put $\chi(x, y) = (-1)^{x+y}$. The quantity

$$\sigma(X) = \begin{cases} \chi(p), & h \text{ is vertical,} \\ -\chi(p), & h \text{ is horizontal} \end{cases}$$

is invariant, since location parity and heading axis both change on each update. We fix $d = (2, -2)$. Lemma 3.2 will derive $\sigma = -1$ and a canonical start at the origin facing east from the prescribed wake.

Definition 2.1 (Complete odd-class wake)**.** Two positions belong to the same drift class if their difference lies in $\mathbb{Z}d$. For $d = (2, -2)$ the complete class label is

$$(t, r) = (x + y,\ x \bmod 2).$$

The wake of a word is the set of classes containing an odd number of its phase positions. The prescribed standard wake is

$$\begin{aligned} O_{\mathrm{std}} = \{&(-10, 0), (-10, 1), (-9, 1), (-8, 1), (-7, 0), (-6, 0),\\ &(-5, 0), (-4, 1), (-2, 0), (-1, 1), (0, 0), (0, 1)\}. \end{aligned} \tag{1}$$

The quotient uses the full drift. Repeating a word twice, for example, doubles its drift and changes the quotient in which its wake is specified.

Theorem 2.2 (Uniqueness under the standard wake)**.** *Every finite-support-realizable, heading-restoring periodic word with drift* $(2, -2)$ *and wake* (1) *is the standard* $P104$ *word. In the normalized phase at the top class* $(0, 0)$ *its period is* 104 *and its word is exactly* w_* *displayed in Appendix B. The corresponding translations, quarter turns, and phase changes give the equivalent geometric statement.*

Corollary 2.3. *If a finite initial state has an eventual periodic suffix with the above drift and complete wake, up to the stated normalization, then its future is eventually the standard highway.*

Proof. The black field at any finite suffix boundary is still a complete finite set. Apply Theorem 2.2 at that boundary. □

The conclusion identifies the read trace. Complete initial fields may differ on cells that the future trace never reads. The twelve class labels in (1) specify both transverse levels and longitudinal residues.

3 Finite-support realization and extreme classes

We recall the chronological criterion of [2, Theorem 3.1 and Remark 3.2]. The proof identifies exactly where finite support is used.

Fix a class and write its phase positions as $q_i = b + a_i d$. The visit to the physical cell $b + nd$ at phase i occurs at time

$$(n - a_i)P + i = nP + (i - Pa_i), \qquad n \geq a_i.$$

Thus stable visit order is increasing $i - Pa_i$, or equivalently decreasing a_i, with increasing i to break ties.

Lemma 3.1 (Supplied-word criterion)**.** *A heading-restoring word of nonzero drift has a finite-support periodic realization if and only if the stable word in every class is* $\mathrm{R}, \mathrm{L}, \mathrm{R}, \mathrm{L}, \ldots$*, starting with* R*. A finite seed can be constructed from the word.*

Proof. For all sufficiently large n, every phase in the class occurs at $b + nd$. All but finitely many such cells are initially white. Their first turn is therefore R, and each successive reading flips the colour.

Conversely, at $b + nd$ the available phases are obtained from the stable order by deleting all groups with $a_i > n$. This deletes a prefix and leaves an alternating suffix. For $\min_i a_i \leq n < \max_i a_i$, place a black cell precisely when the number of deleted phases is odd. Put no other black cells in this class. The suffix then starts with the required colour. There are only finitely many boundary cells and classes. Induction over literal time proves execution of every prescribed update. □

Consequently each class has signed excess $\#\mathrm{R} - \#\mathrm{L}$ equal to its indicator in the odd wake. This supplies the exact balances used below.

Lemma 3.2 (Extremal singleton classes)**.** *For a realization with wake* (1)*, $\sigma = -1$ and every phase position satisfies $-10 \le t \le 0$. Every class visited on either extreme level has exactly one event, a right turn. Top events arrive heading east; bottom events arrive west. There is a unique event of the top class* $(0, 0)$*.*

Proof. At the highest visited level there is no predecessor or successor at a higher level. Arrival and departure directions are therefore forced by the axis invariant. Every event of an extreme class has the same turn. Lemma 3.1 permits only a singleton right turn in that class, so the extreme class must be odd. The same argument applies at the lowest level. Since the extreme levels of the given odd wake are -10 and 0, they are also the trace extremes. At the top level $t = 0$, a right turn that both arrives from and returns to the lower level must enter eastward and leave southward. Hence $\sigma = -1$. The bottom right turn enters westward and leaves northward. Both residue classes at each extreme belong to (1), and hence occur exactly once. □

A top event in class $(0, 0)$ has position $(2k, -2k)$. Changing the phase to that event and translating by $-kd$ preserves every class label in (1). This gives the canonical phase in Theorem 2.2. Choosing the other top event as origin gives the equivalent description with both residues swapped.

4 A finite certificate for an all-period bound

The upper bound uses a finite graph that retains transverse position, heading and a short turn context. Every actual periodic trace lifts to a closed walk in this graph. Its integer edge inequalities will then bound the length of every such trace.

4.1 Forbidden factors and geometric context

If a proposed finite factor reads one physical cell twice consecutively with the same turn, the factor cannot occur on any initial colouring. Generate physically consistent first-return words of at most 20 updates, append the initial turn again, and include complemented words. The resulting 588 forbidden factors have length at most 21. Each is separately checked by geometric reconstruction; completeness of this list is unnecessary.

The contexts are the empty word and the proper viable prefix contexts of these factors. After a new letter, retain the longest suffix that is again a context, rejecting a forbidden suffix. There are 4,931 contexts. A vertex is (t, h, a): level, heading, and context. Retain only headings compatible with $\sigma = -1$ and contexts whose entire geometric placement lies in $-10 \le t \le 0$. The latter placement is recovered by reversing the net displacement of the context and rotating to its current heading. A graph edge is one literal turn and unit move with the corresponding context update.

The regenerated graph has 54,718 vertices and 90,158 edges. After context warmup, a periodic trajectory lifts to a length-P closed walk: its transverse position, heading and suffix context repeat after one period. Vertices and edges may be revisited.

Lemma 4.1 (Degree peeling preserves every closed walk)**.** *Repeatedly delete a vertex with zero incoming or zero outgoing degree in the currently retained graph, together with its incident edges. Every closed walk in the original graph remains in the final graph.*

Proof. If a vertex of a closed walk were deleted, consider the first deleted vertex on that walk. Its predecessor and successor would still be present, so it would still have an incoming and an outgoing edge, a contradiction. □

This reduction retains 24,424 vertices and 37,330 edges. A separate reverse-insertion check verifies the deletion order. We verify the upper potential on this smaller graph; Lemma 4.1 preserves the coverage needed for the period bound.

4.2 An exact integer potential

For levels $t = -10, \dots, 0$ define

$$\alpha_t = -2(-1)^t(t+5)^2,$$

that is,

$$\alpha = (-50, 32, -18, 8, -2, 0, -2, 8, -18, 32, -50).$$

For an edge $v \to w$ with source level t, put $\varepsilon = 1$ for R and -1 for L.

Lemma 4.2 (Finite upper certificate)**.** *The regenerated graph admits integers $0 \le y_v \le 180$ such that on every edge*

$$y_v - y_w + \alpha_t \varepsilon \le -1. \tag{2}$$

Computer-assisted proof. Initialize all y_v to zero and repeatedly enforce $y_w \ge y_v + 1 + \alpha_t \varepsilon$ along queued outgoing edges. Abort, rather than accept, if any value exceeds 180. With this cap the construction terminates: every change raises an integer, and there are at most $180|V|$ changes. Once the queue is empty, a separate scan verifies (2) on every edge, without relying on the construction algorithm. The supplement regenerates the reduced graph and its potential, records the construction statistics, and checks each retained inequality using integer arithmetic. The reconstructed potential ranges from 0 to 129, and the maximum left-hand side is -1. □

Proposition 4.3 (All-period upper bound)**.** *Every realization in Theorem 2.2 has $P \le 168$.*

Proof. The per-level odd counts are

$$o = (2, 1, 1, 1, 1, 1, 1, 0, 1, 1, 2).$$

Summing (2) on the lifted closed walk cancels the vertex terms. Lemma 3.1 identifies the signed turn excess on layer t with o_t. Direct integer arithmetic gives $\sum_t \alpha_t o_t = -168$. Thus $-168 \le -P$. □

The upper graph depends only on the per-level counts. The next stage uses the full class labels in (1) to determine which bounded walks are realizable.

5 An exact ordered-event representation

In each complete class (t, r) let m be its number of events and index them by stable rank $k = 0, \dots, m-1$. The turn is right at even rank and left at odd rank. Let $a_k = 1$ indicate arrival from $t-1$, and $d_k = 1$ indicate departure to $t-1$. The literal rule is

$$d_k = \begin{cases} a_k \text{ xor } (k \bmod 2), & t \text{ even}, \\ 1 - (a_k \text{ xor } (k \bmod 2)), & t \text{ odd}. \end{cases}$$

At even levels arrival is east or west according as $a_k = 1$ or 0; at odd levels it is north or south.

Every directed lattice edge has one direction of use in the fixed checker sector. Its source occurrences and target occurrences match in stable order, first to first and so on: physical time increases by one on that edge, and passage to the target representative changes lifted time by a constant. There is no free matching permutation.

Use representatives $b_{t,r} = (r, t-r)$. An edge has integer voltage δ determined by

$$b_{\text{source}} + e_{\text{outgoing}} = b_{\text{target}} + \delta d.$$

A vertical edge has voltage zero. East moves from residue 1 to 0 have voltage 1; west moves from 0 to 1 have voltage -1. Other horizontal edges have voltage zero.

Proposition 5.1 (Exact event-graph test)**.** *A complete matched graph represents a normalized realization if and only if it forms one directed cycle of total voltage* 1*, and the lifted times $j_i - Pv_i$, read along that cycle from the top event* $(0,0)$*, strictly increase with rank in every class. Here j_i is cycle index and v_i is accumulated voltage.*

Proof. A real trajectory gives these matches and times, so necessity follows. Conversely, following the proposed cycle reconstructs a literal walk with the correct headings and drift d. The lifted-time test makes its stable class order exactly the rank order. The turn assigned to every rank therefore satisfies Lemma 3.1, which constructs the finite seed. The explicit seed is then checked independently as described in Section 7. □

In the normalized gauge each word determines its event graph. Enumerating all such graphs therefore enumerates words, not merely a selection of seeds.

6 Exhaustion without enumerating all binary words

The search builds complete classes one transverse layer at a time. It enumerates ordered arrival strings, joins forced edge occurrences immediately, and rejects partial graphs by necessary conditions. A completed traversal supplies the finite classification.

6.1 Exact count completion and safe merging of count histories

Suppose a new class has I arrivals and J departures on the already processed side. Let $o = \mathbf{1}_{O_{\text{std}}}(t,r)$ and let a count processed-side arrivals at even ranks. Complement both side bits when the construction runs downward. The possible class lengths are exactly

$$m = \begin{cases} 2(I+J-2a)+o, & t \text{ even}, \\ 2(J-I+2a)-o, & t \text{ odd}, \end{cases} \qquad 0 \le a \le \lceil m/2 \rceil, \quad 0 \le I-a \le \lfloor m/2 \rfloor. \tag{3}$$

These formulas follow by counting the outgoing bits in the local rule. Conversely, the inequalities choose the required number of ones at even and odd ranks, so they are an exact count feasibility test.

A boundary stores its two outgoing and two incoming counts. Transitions between boundaries retain the residue swap on horizontal edges. A dynamic program merges count histories with the same boundary, keeping the smallest completion cost. Every future count transition depends only on that boundary, so the minimum gives a lower bound on the remaining number of events. The event search separately retains all forced matches and chronological constraints.

Both directions of this count program show a minimum total cost 104. At cutoff 104 the top-down boundary-state counts are

$$1, 1, 6, 81, 174, 92, 65, 19, 8, 1.$$

The required boundary just above the bottom has minimum cost 102; the two bottom singletons add 2. Any putative period below 104 has all partial nonnegative costs below 104 and is therefore covered by this finite calculation. Twelve minimum count profiles remain for the exact chronological search.

6.2 Phase congruence

Induction in (3) fixes the class-length residues modulo four. From bottom to top they are

$$(1,1),(0,3),(2,3),(1,2),(1,0),(3,0),(2,3),(2,2),(3,2),(0,3),(1,1).$$

For the normalized physical walk,

$$i \bmod 4 = 2(x_i \bmod 2) + (x_i + y_i \bmod 2).$$

This follows directly by checking the four update phases starting east at the origin. A heading-restoring period with horizontal displacement 2 has an even number $P/2$ of horizontal moves, so $4 \mid P$. Each of the four phase groups contains $P/4$ events. Their displayed residue sums are $(10, 6, 10, 10)$, all 2 mod 4. Hence

$$P \equiv 8 \pmod{16}. \tag{4}$$

Together with the upper bound and the count minimum, the only candidate lengths are $104, 120, 136, 152, 168$.

6.3 Chronological difference constraints

A partial graph that closes a directed component before the final layer cannot be completed into one cycle: required future events could no longer join that fixed component. All other components are directed open chains.

On one chain write

$$\tau_u = z_{\mathrm{chain}(u)} + p_u - Pv_u,$$

where p_u is its position in the chain and v_u its accumulated voltage. Consecutive ranks in a class have the same phase modulo four. Their strict stable order therefore requires a gap at least four:

$$z_{\mathrm{chain}(v)} \geq z_{\mathrm{chain}(u)} + 4 + p_u - p_v + (v_v - v_u)P. \tag{5}$$

A positive cycle of these difference constraints is impossible.

The final period is not yet known. Let n be the number of constructed events and L the search cap. It is safe to test (5) at $P = L$, as the following argument shows. For a simple cycle on q chain variables, its total weight is $C + KP$, where $K \in \mathbb{Z}$. Each distinct chain contributes one difference of two indices. Its index range is its length minus one, and the q chains together contain at most n vertices. Thus

$$5q - n \leq C \leq n + 3q.$$

If $K \geq 1$, the cycle is positive for every $P \geq n$. If $K = 0$, its sign is independent of P. If $K \leq -1$, its weight decreases with P. A positive cycle at L consequently rules out every $n \leq P \leq L$. This proves that evaluating the constraints at the search cap is a sound pruning rule. All checks use exact signed integers.

6.4 A local refinement: consecutive ranks on a fixed chain

A fixed chain of matched edges is already a contiguous piece of physical motion. Its repeated visits to a cell must therefore occupy consecutive positions in that cell's ordered visit list.

Lemma 6.1 (Consecutive-visit rank pruning)**.** *Let an open chain of fixed edges be part of a realizable event graph. Lift it to the lattice by accumulating edge voltages. If two consecutive occurrences of one physical lattice cell in this chain have stable class ranks k and ℓ, then $\ell = k + 1$.*

Proof. Translate a sufficiently late occurrence of the chain along the periodic trajectory. Every class involved is then in the stable finite-support regime. The chain is a contiguous sequence of actual updates; its length does not exceed the final period. Between the stated occurrences there is no intervening visit to that physical cell. They are therefore consecutive in the cell's stable visit list. Its list is indexed by consecutive ranks $0, \dots, m-1$, proving the assertion. □

In particular, equal turn bits are forbidden, but mere bit alternation is weaker: a proposed jump from rank 0 to rank 3 would alternate turns while omitting required visits.

For implementation, a node of label (t, r) at accumulated voltage v has lifted coordinate $(r + 2v, t - r - 2v)$. A stamped array records the previous rank at each cell while traversing one open chain. Separate chains receive separate stamps; they are not assumed to share an absolute placement. This check is inserted after premature closed cycles are rejected and before the more expensive difference-constraint solver.

6.5 Separation of fixed trajectory chains

Open chains also constrain each other's possible placement in time. Each is a contiguous block of updates, and distinct chains contain disjoint phase events. Their copies in a completed periodic trajectory therefore occupy disjoint time intervals.

Lemma 6.2 (Chain separation)**.** *Suppose consecutive ranks of a class lie at events u and v in distinct open chains A and B. Let ℓ_A be the number of events in A. Their chronological constraint strengthens to*

$$z_B \geq z_A + \max\{4 + p_u - p_v, \ell_A\} + (v_v - v_u)P. \tag{6}$$

Proof. Shift sufficiently late copies of the two chains by whole periods so that u and v refer to the same physical cell. Their respective root times are $z_A - Pv_u$ and $z_B - Pv_v$, up to a common epoch offset. Stable rank order places u before v. Since the two contiguous chain intervals are disjoint, the whole interval for A precedes that for B. Thus $z_B - Pv_v \geq z_A - Pv_u + \ell_A$. Combining this inequality with (5) proves the claim. □

The stronger constraint is especially convenient at an unknown period. Each interchain edge has weight $C_e + K_e P$ with $C_e \geq \ell_A \geq 1$. A cycle has weight $C + KP$ with $C > 0$. If $K \geq 0$, it is positive for every positive P; if $K < 0$, it increases as P decreases. A positive cycle at the search cap therefore excludes all smaller candidate periods. This strengthening uses the existing constraint graph and introduces no additional enumeration dimension.

6.6 Completed finite classification

The original, rank-pruned and chain-separated searches all complete the cap 168, returning exactly the word w_* of Appendix B. Their internal prefix counts are:

cap	original	consecutive ranks	chain separation
104	1,692	1,443	1,376
120	26,874	18,644	17,566
136	562,700	313,711	270,080
152	5,781,232	2,164,812	1,739,460
168	40,227,976	12,193,595	8,580,751

Each row completely searches every period not exceeding the cap. Every run accepts the same single normalized 104-letter word. At cap 168, chain separation reduces the rank-pruned prefix count by 29.63%, and the two rules together reduce the original count by 78.67%. A prefix is an internal enumeration state: it records a partial choice of class arrival strings and their forced matches.

The supplement retains all three enumerators as a fixed source and small source-pinned overlays. Complete runs compare their accepted word lists.

Proof of Theorem 2.2. By Proposition 4.3, every candidate has period at most 168. Lemma 3.2 places it in the normalized event domain. Proposition 5.1 and the complete enumeration cover every candidate in that domain. The count, cycle, and chronology rejections above cannot remove a realization. The complete output contains only w_*. The independently checked finite seed realizes w_*^ω and identifies it with a phase of the standard word. Thus existence and uniqueness both hold. The two pruning refinements supply further completed traversals of the same domain with the same unique output. □

7 Exact verification and reproducibility

The proof combines written coverage arguments with exact finite computations. The upper-bound checker regenerates the context graph and verifies every integer edge inequality. The classification checker regenerates the count completion tables, enumerates the ordered-event graphs and requires a completed traversal. Source hashes and deterministic reports identify the inputs and outputs of both stages.

For a supplied seed B and period word, an independent all-future verifier computes the read bit at phase i, epoch k as

$$\mathbf{1}_B(q_i + kd) \ \text{ xor } \ (\#\{(j,s) : q_j - q_i = sd,\ s > 0 \text{ or } (s = 0, j < i),\ s \le k\} \bmod 2)\,.$$

The seed term has finitely many hits, and the historical-write term changes at finitely many thresholds. Checking zero, every threshold, every hit, and one epoch after each hit checks every constant interval, including the final infinite interval. This formula checks the full seed, including any black cells placed far along the future path.

Controls include all rotations of the accepted seed, deletion of its active cells, a far-ahead blocker, and a deliberately interrupted search that must report a resource limit. Independent geometric controls test the rank and chain-separation rules, including decompositions of the genuine period into open chains and an explicit pair of chains whose rank orders would force overlapping intervals. The general budget checks cover exact local identities, all 104 phases and four rotations of the standard word, and its first three powers with their corresponding drifts and wakes. A complete small inverse problem is cross-checked against all 32,766 nonempty words through length 14.

The self-contained source and retained reports are in the wake synthesis project. From the repository root,

```
python3 wake_synthesis/reproduce.py
```

runs the complete uniqueness replay and the general algebraic checks. Adding `--baseline` also repeats the original cap-168 enumeration; `--search` runs the supplementary finite detour experiment. Python 3.10 or later and a C++17 compiler suffice. The computational acceptance conditions use exact integer or rational arithmetic. Build the manuscript with `make -C wake_synthesis/paper`.

8 Conclusion

The complete standard wake is rigid: together with its full drift, it uniquely determines the classical 104-step highway among finite-support periodic realizations. Local forbidden factors supply a universal period budget, and ordered class visits recover the only possible motion inside that budget. The general wake bounds extend the inverse-synthesis question to every non-axis drift. Further work can apply these methods to other complete wakes, compress the finite certificates, and connect wake constraints with the evolution from arbitrary finite initial states.

AI assistance. Generative AI assisted with developing and checking arguments, implementing verification programs, and revising the exposition.

A Wake-dependent period bounds and inverse synthesis

The standard-wake classification is one instance of a general inverse problem. Prescribe a non-axis drift d, an initial heading, and a finite set $O \subset \mathbb{Z}^2/\mathbb{Z}d$ of odd classes. We seek a periodic word and finite seed with exactly these data. The following bounds, developed in the first version [5], determine a finite search range directly from the input wake.

Lemma 3.1 gives $\#\mathrm{R} - \#\mathrm{L} = |O| =: g$. Heading restoration implies $g \equiv 0 \pmod 4$, P even and $d_x + d_y$ even. For $d = (m, -m)$, the complete class label is $(x + y, x \bmod m)$. The checker-sector invariant σ is determined by the supplied pose.

A.1 An integer clock for diagonal budgets

After rotating a diagonal drift parallel to $(1, -1)$, the transverse coordinate $x+y$ and heading return after each period. A clock using these quantities has a cancelling head correction: its period gain comes from the wake. We recall the unit increment identity of [4] before computing that gain.

Fix an integer c, set $s = x + y - c$, and put

$$f(s) = 3^{|s|} - 1, \qquad \epsilon(h) = \begin{cases} +1, & h = N, E, \\ -1, & h = S, W, \end{cases} \qquad G(s, e) = \begin{cases} \frac{1}{2} f(s), & es \geq 0, \\ -\frac{3}{2} f(s), & es < 0. \end{cases}$$

Use the integer clock of [4] on the complete finite state

$$\Theta_c(X) = -\sigma(X) \sum_{q \in B} \chi(q) f(q_x + q_y - c) + G(p_x + p_y - c, \epsilon(h)). \tag{7}$$

Since $f(s)$ is even, Θ_c is an exactly computable integer.

Lemma A.1 (Integer clock, Ke [4])**.** *For every finite-support state and integer centre c, with e, e' the before and after values of ϵ and $s = p_x + p_y - c$,*

$$\Theta_c(FX) - \Theta_c(X) = 1 + 4 f(s) \mathbf{1}_{\{e' \neq e,\ es < 0\}} \geq 1. \tag{8}$$

Proof. Write $\delta = +1$ on R and -1 on L. Directly checking the two colours at four headings gives $\delta \sigma \chi(p) = ee'$, and the outgoing step gives $s' = s + e'$. Thus the black-field increment is $-ee' f(s)$. Let $G_\pm(s) = G(s, \pm 1)$. The elementary recurrences are

$$G_+(s+1) - G_+(s) = f(s) + 1, \qquad G_-(s-1) - G_-(s) = f(s) + 1,$$
$$G_-(s) - G_+(s) = -2 \operatorname{sgn}(s) f(s).$$

If $e' = e$ the total gain is one. For $+1 \to -1$ it is $1 + 2f(s) + G_-(s) - G_+(s)$, equal to one for $s \geq 0$ and to $1 + 4f(s)$ for $s < 0$. The other reversal gives the reflected cases. This includes $s = 0$. □

A.2 Diagonal wake budgets and crossing refinement

Rotate a diagonal drift to $d = (m, -m)$, $m > 0$, and use the transformed heading. Let $t = x + y$, let o_t count odd classes at level t, and define

$$A_t = -\sigma(-1)^t o_t, \qquad K_c(O) = \sum_t A_t(3^{|t-c|} - 1). \tag{9}$$

Theorem A.2 (Wake-determined diagonal period)**.** *Every realization of this wake obeys $P \le K_c(O)$ for every integer c.*

Proof. Over one period the head term of (7) cancels, since both t and the heading return to their initial values. The signed cell toggles give $-\sigma \sum_i \chi(q_i) f(t_i - c)\delta_i$. The weight is constant in each full d-class. Lemma 3.1 makes its net δ count one on odd classes and zero on even classes. The exact period increment is therefore $K_c(O)$. Summing (8) proves the bound. The unchanged part of the full field cancels algebraically; it is not deleted. □

Here is an independent geometric derivation, which also supplies a refinement. The finite projected period is a closed walk with increments ± 1 in t. Let f_t count upward crossings of the cut $(t, t+1)$; it also counts downward crossings. Let m_t count visits to t, and u_t count passages entering and leaving in the increasing direction. Flow conservation equates this with the number of decreasing-through passages. The fixed arrival axis and the two turn choices give

$$m_t = f_{t-1} + f_t = A_t + 4u_t, \qquad 0 \le u_t \le \min(f_{t-1}, f_t). \tag{10}$$

For example, at vertical-arrival levels, right turns are through passages and left turns reverse the projection, giving $\#\mathrm{R} - \#\mathrm{L} = 4u_t - m_t$; horizontal arrival reverses that sign. Substituting $u_t \le f_{t-1}$ into $f_t = A_t + 4u_t - f_{t-1}$ gives the first inequality below; $u_t \le f_t$ gives the second:

$$f_t \le 3f_{t-1} + A_t, \qquad f_{t-1} \le 3f_t + A_t. \tag{11}$$

Let ℓ, h be the extreme wake levels. Outside them $A_t = 0$; starting past the finite support of the projected walk and applying (11) inward forces $f_{\ell-1} = f_h = 0$. Thus the periodic trace lies between the extreme wake levels. An empty wake forces every $f_t = 0$, impossible for a nonempty period.

Starting from these two zero boundary values, set

$$L_{\ell-1} = 0, \quad L_t = 3L_{t-1} + A_t, \qquad R_h = 0, \quad R_t = 3R_{t+1} + A_{t+1}, \quad C_t = \min(L_t, R_t).$$

Then $0 \le f_t \le C_t$ for $\ell \le t < h$. If a C_t is negative, the wake is impossible; otherwise

$$P = 2\sum_t f_t \le 2\sum_t C_t. \tag{12}$$

Moreover $d_x = \sigma \sum_t (-1)^t f_t$: on a cut with $\sigma(-1)^t = 1$, the upward horizontal moves contribute $+f_t$ to x; on the other cuts the downward horizontal moves contribute $-f_t$. If C_+ and C_- sum the corresponding positive and negative cut bounds, then also

$$P \le 4C_+ - 2d_x, \qquad P \le 4C_- + 2d_x. \tag{13}$$

Corollary A.3 (Width and growth bound)**.** *Writing $W = h - \ell$, a diagonal realization has even W and*

$$P \le g(3^{W/2} - 1). \tag{14}$$

Proof. At a transverse extreme every passage reverses the projection. The fixed arrival axis makes all letters in an extremal class equal. Lemma 3.1 therefore forces exactly one R per such class. Both extremes have the same arrival axis, so W is even; this is the usual extremal rigidity argument. Take $c = (\ell + h)/2$ in Theorem A.2, discard negative coefficients and use $\sum_t \max(A_t, 0) \le g$ to obtain (14). □

For the normalized standard wake, $\sigma = -1$ and the coefficients at $t = -10, \dots, 0$ are

$$A = (2, -1, 1, -1, 1, -1, 1, 0, 1, -1, 2).$$

The cut bounds are $C = (2, 5, 16, 47, 142, 145, 48, 16, 5, 2)$, with sum 428. Thus (12) gives $P \le 856$, as does (9) with centre $c = -5$. The chronological constraints in Section 4 strengthen this budget to the cutoff used in the complete classification.

A.3 General non-axis drift by reciprocal moments

For a general drift, use a transverse linear coordinate constant along it. An exponential weight and its reciprocal give complementary period budgets; Cauchy–Schwarz then bounds the number of steps independently of their individual weights.

Rotate the drift to $(a_0, -b_0)$, $a_0, b_0 > 0$. Put $k = \gcd(a_0, b_0)$, $\eta = (b_0/k, a_0/k)$, and choose rational $r > 1$ such that, with $a = r^{\eta_x}$ and $b = r^{\eta_y}$,

$$\kappa = 1 - \frac{2}{a+1} - \frac{2}{b+1} > 0.$$

The choice $r = 4$ always works. Define $w(q) = r^{\eta \cdot q}$, and

$$(\gamma_N, \gamma_E, \gamma_S, \gamma_W) = \left(\frac{2}{b+1}, \frac{2}{a+1}, -\frac{2b}{b+1}, -\frac{2a}{a+1} \right).$$

The one-sided potentials are

$$\Psi_+(X) = -\sigma \sum_{q \in B} \chi(q) w(q) + \gamma_h w(p), \qquad \Psi_-(X) = -\sigma \sum_{q \in B} \frac{\chi(q)}{w(q)} + \frac{\gamma_{h+2}}{w(p)}.$$

Substitution of the current-cell flip gives $\Delta\Psi_+ = \kappa w(p)$ in six heading/colour cases and $(\kappa + 4)w(p)$ in S/L and W/R. The half-turn formula gives $\Delta\Psi_- \ge \kappa / w(p)$. These local identities are the one-sided exponential clock construction, included to make the new period argument self-contained.

Define the two finite wake moments, using any representatives of O,

$$A_+ = -\sigma \sum_{q \in O} \chi(q) r^{\eta \cdot q}, \qquad A_- = -\sigma \sum_{q \in O} \chi(q) r^{-\eta \cdot q}.$$

They are independent of the representatives since $\eta \cdot d = 0$ and $d_x + d_y$ is even. The head terms cancel over a period, and Lemma 3.1 gives exact increments A_+ and A_-. Thus both are positive and $\kappa \sum_i w(q_i) \le A_+$, $\kappa \sum_i w(q_i)^{-1} \le A_-$. Cauchy–Schwarz yields:

Theorem A.4 (Reciprocal wake bound)**.** *Every realization with the specified non-axis drift and wake satisfies*

$$P^2 \le \frac{A_+ A_-}{\kappa^2}. \tag{15}$$

All quantities are rational. The integer square root of the floor of the right side is a computable cutoff. Changing the transverse centre scales A_+ and A_- reciprocally, leaving the product unchanged. For axis drift the normal has a zero coordinate, so this construction cannot give positive κ.

A.4 All-period synthesis and detour obstruction

Theorem A.5 (Period-unknown inverse synthesis)**.** *Given a non-axis integer drift, an initial heading and a finite odd-class wake, existence of a finite-support periodic realization is decidable. A positive answer can produce its word and an explicit finite seed.*

Proof. First reject odd $d_x + d_y$, growth $|O|$ not divisible by four, or a nonpositive moment A_+ or A_-. Otherwise compute Theorem A.4's cutoff, additionally using Theorem A.2 and (12)–(13) in the diagonal case. A negative crossing bound also rejects the input. Enumerate every nonempty word up to the least derived bound. Check its heading and drift, apply Lemma 3.1, and require the exact odd-class set. A passing word constructs a finite seed; complete finite exhaustion gives a negative answer. Completeness follows from the period bound. □

The all-future verifier of Section 7 independently checks the constructed finite seed. For cutoff B, enumeration visits at most $2^{B+1} - 2$ nonempty binary words; the period bound guarantees termination.

Corollary A.6 (No wake-preserving closed-pose insertion)**.** *Suppose realizable words u, v have the same non-axis drift, and v is obtained by inserting nonempty detours into u, each returning to the same position and heading while leaving the original steps unchanged. Then*

$$A_+(O_v) > A_+(O_u), \qquad A_-(O_v) > A_-(O_u).$$

Thus their signed transverse wake cannot be identical. For diagonal drift, $K_c(O_v) - K_c(O_u) \geq |v| - |u| > 0$ for every integer c.

Proof. Every retained original step has the same position, heading and letter; hence it has the same local clock increment. Every inserted step has strictly positive increment in both one-sided clocks, and at least one in the integer clock. Sum over the words and use the exact wake-increment equalities. □

Thus any such realizable detour requires an additional wake budget.

B The normalized word and a full finite witness

The unique word is the concatenation of the following four strings:

```
RRRLRLLRRLLLLRRLLRRRRLLRRL
RLLRLLRLRLLRRRRLLRLLRRRRLL
RRRRLLRLRRRRLRLLLLRRRRLRRL
RRRRLLLLRLRRRRLRRRRLLLLRLR
```

The four strings are concatenated without spaces and are checked against the machine-readable witness. Its period is 104, with 58 right turns, 46 left turns, and drift $(2, -2)$. A complete finite seed, at the origin facing east, is

$$\begin{aligned} B_* = \{&(-7,0),(-7,1),(-6,-3),(-6,-2),(-6,-1),(-6,2),\\ &(-5,-2),(-5,0),(-5,1),(-5,2),(-4,-2),(-4,1),\\ &(-3,-3),(-3,1),(-2,-3),(-2,-1),(-2,1),(-1,-2),(-1,0)\}. \end{aligned}$$

The exact per-class counts, bottom to top, are

$$(1,1),(4,3),(6,3),(5,6),(5,8),(7,8),(10,7),(6,6),(3,6),(4,3),(1,1).$$

The seed realizes the displayed trace from time zero, as certified by the all-future breakpoint check.

C Certificate reconstruction specification

To make the finite upper lemma reproducible independently of a numerical optimizer, the supplement specifies the following deterministic procedure. Use bits 0 and 1 for right and left. Branch on the first read of a new cell, and otherwise follow the colour left by its previous read. Explore first returns to the starting cell of length at most 20, pruning only when the remaining unit moves cannot reach the start. Append the original starting bit at the return, include complements, and sort and deduplicate the resulting factors. Every proposed factor is then rechecked for an equal-bit consecutive reread.

Insert all factor prefixes into a trie, remove forbidden contexts, and use longest-suffix transitions. Place each retained context geometrically relative to its endpoint for each heading. Intersect these placements with the derived transverse interval and checker-sector constraint. This produces the full graph of Section 4. Perform the degree-peeling reduction, regenerate the capped integer potential, and rescan every retained edge.

For the exact stage, regenerate the opposite-direction minimum-cost tables from (3). Enumerate all arrival strings compatible with the row margins, perform the fixed stable edge match, and apply the exact rejection rules in Section 6. At the bottom, add the two singleton events before testing for one final cycle; a legitimate final cycle must not be confused with a prematurely closed component.

The supplementary manifest records every required source hash, accepted word, node count, and finite check result. The original, rank-pruned and chain-separated complete enumerations are reproducible.

References

[1] A. Gajardo. A symbolic projection of Langton's ant. *Discrete Mathematics & Theoretical Computer Science Proceedings*, AB, Discrete Models for Complex Systems (DMCS'03), 2003. doi:10.46298/dmtcs.2312.

[2] A. Jillhewar. *Finite-Support Periodic Highways of Langton's Ant: Necessary Conditions, Transverse Exclusions, and Exact Search*, preprint, 2026. doi:10.21203/rs.3.rs-10516269/v1. Updated source inspected at revision `76a7e234e577`.

[3] A. Jillhewar. *Width-four research log*, 21 July 2026, revision `76a7e234e577`.

[4] H. Ke. *An Ant on the Clock: Exponential Escape Bounds for Langton's Ant*, Zenodo, version 0.4.0, 2026. doi:10.5281/zenodo.23402379.

[5] H. Ke. *Reading the Wake: Period Bounds and Inverse Synthesis for Langton's Ant*, Zenodo, version 0.1.0, 2026. Previous version of this preprint. doi:10.5281/zenodo.23402422.